

■ 3.4.6 Eliminating Variables

When you write down a set of simultaneous equations in *Mathematica*, you are specifying a collection of constraints between variables. When you use `Solve`, you are finding values for some of the variables in terms of others, subject to the constraints represented by the equations.

<code>Solve[eqns, vars, elims]</code>	find solutions for <i>vars</i> , eliminating the variables <i>elim</i> s
<code>Eliminate[eqns, elims]</code>	rearrange equations to eliminate the variables <i>elim</i> s

Eliminating variables.

Here are two equations involving *x*, *y* and the “parameter” *a*.

```
In[1]:= eqn = {x == 1 + 2 a, y == 9 + 2 x}
Out[1]= {x == 1 + 2 a, y == 9 + 2 x}
```

If you solve for both *x* and *y*, you get results in terms of *a*.

```
In[2]:= Solve[eqn, {x, y}]
Out[2]= {{x -> 1 + 2 a, y -> 9 + 2 (1 + 2 a)}}
```

Similarly, if you solve for *x* and *a*, you get results in terms of *y*.

```
In[3]:= Solve[eqn, {x, a}]
Out[3]= {{x -> (-9 + y)/2, a -> -(1/2) + (-9 + y)/4}}
```

If you only want to solve for *x*, however, you have to specify whether you want to eliminate *y* or *a*. This eliminates *y*, and so gives the result in terms of *a*.

```
In[4]:= Solve[eqn, x, y]
Out[4]= {{x -> 1 + 2 a}}
```

If you eliminate *a*, then you get a result in terms of *y*.

```
In[5]:= Solve[eqn, x, a]
Out[5]= {{x -> (-9 + y)/2}}
```

In some cases, you may want to construct explicitly equations in which variables have been eliminated. You can do this using `Eliminate`.

This combines the two equations in the list *eqn*, by eliminating the variable *a*.

```
In[6]:= Eliminate[eqn, a]
Out[6]= y == 9 + 2 x
```

This is what you get if you eliminate *y* instead of *a*.

```
In[7]:= Eliminate[eqn, y]
Out[7]= 2 a == -1 + x
```

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As a more sophisticated example of **Eliminate**, consider the problem of writing $x^5 + y^5$ in terms of the “symmetric polynomials” $x + y$ and xy .

To solve the problem, we simply have to write **f** in terms of **a** and **b**, eliminating the original variables **x** and **y**.

```
In[8]:= Eliminate[ {f == x^5 + y^5, a == x + y, b == x y},
                  {x, y} ]
```

```
Out[8]= -5 a^3 b + 5 a b^2 == -a^5 + f
```